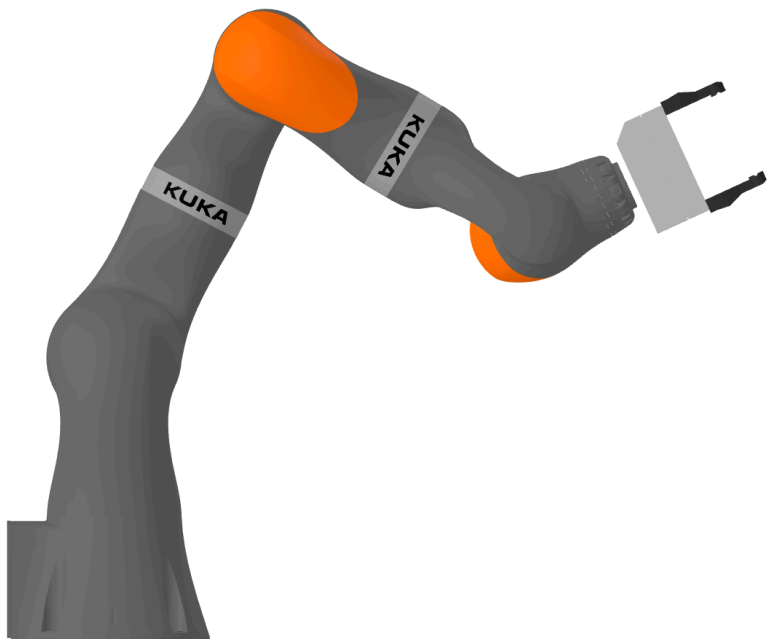
Statics and Grasp Selection



MIT 6.4210/2

Robotic Manipulation

Fall 2026, Lecture 7

Massachusetts Institute of Technology

Credits

Analytical grasping	Stanford **CS237B** (Winter 2025) — Bohg, Pavone, Sadigh `https://web.stanford.edu/class/cs237b/pdfs/lecture/cs237b_lecture_7.pdf` friction cones, grasp wrench space and hull, ϵ and volume metrics
Closure, learned grasping	Jeannette **Bohg**, *Principles of Robot Autonomy II* (AA 274B), L8 and L9, Stanford C-space contacts, force-direction closure, the third contact, Dex-Net, Contact-GraspNet
Course text	*Robotic Manipulation*, Ch. 5 — Russ **Tedrake** `https://manipulation.csail.mit.edu/clutter.html`
Hydroelastic contact	Drake — Damrong **Guoy**, Sean **Curtis**, Rick **Cory**, Alejandro **Castro** and others
Grasping in clutter	Lucas **Manuelli** (2017–2018) · *Grasp Pose Detection in Dense Clutter*, ten Pas & Platt
Systems	Dex-Net: Mahler et al., RSS 2017 · arm farm: Levine et al. · Contact-GraspNet: Sundermeyer et al., ICRA 2021 · GraspGen: Murali et al., NVIDIA 2025

A grasp

an act of restraining an object's motion
through application of forces and torques
at a set of contact points

Given an object, where do I put the fingers,
and will the object stay there?

Three ways to answer it

	needs	gives
Analytical	geometry, contacts, μ, mass	a certificate
Heuristic	surface geometry (point cloud)	a candidate, fast
Learned	data — simulated or real	ranked candidates, novel objects

a trade between what you must know and what you can promise

Part I

analytical

Spatial force = wrench

$$F^{B_p}_{\text{name},\,C} = \begin{bmatrix} \tau_C^{B_p} \\ f_C^{B_p} \end{bmatrix} \qquad \begin{matrix} 3 \times 1 \text{ rotational} \\ 3 \times 1 \text{ translational} \end{matrix}$$

B_p: the point on body B where it acts C: expressed-in frame

rotational on top — as in ${}^{A}V_C^B = [\,{}^{A}\omega_C^B;\ {}^{A}\mathrm{v}_C^B]$, and as in Drake

Two operations

re-express — blockwise, one rotation per half

$$ {}^{B}R^{C}\, F_{C}^{B_p} \;\equiv\; \mathrm{diag}({}^{B}R^{C},\, {}^{B}R^{C})\, F_{C}^{B_p} $$

shift the application point — forces are real, torque is a summary

$$ f_{i,B}^{B} = f_{i,B}^{B_{c_i}}, \qquad \tau_{i,B}^{B} = \tau_{i,B}^{B_{c_i}} + {}^{B}p^{B_{c_i}} \times f_{i,B}^{B_{c_i}} $$

static equilibrium is then addition — once everything shares a frame and a point

The cross product, as a multiply

$$d \times f = S(d)\, f, \qquad S(d) = \begin{bmatrix} 0 & -d_z & d_y \\ d_z & 0 & -d_x \\ -d_y & d_x & 0 \end{bmatrix}$$

only the part of f perpendicular to the lever arm turns the object

(the same S as the Jacobian construction in Lecture 4)

Static equilibrium

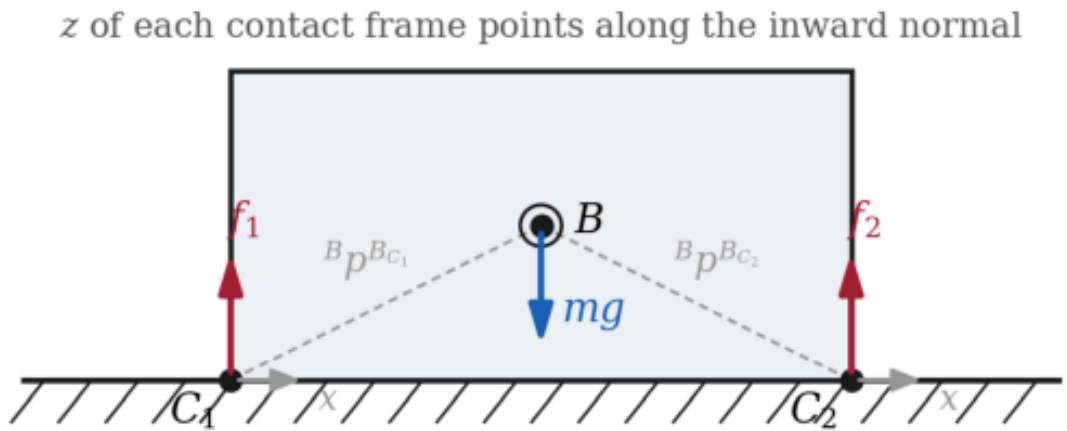
express every wrench in the same frame, shift it to the same point, and add

$$\sum_i F^B_{i,B} + F^B_{\text{ext},B} = 0$$

$$\underbrace{\sum_i \left(\tau^{B_{c_i}}_{i,B} + {}^B p^{B_{c_i}} \times f_i\right) + \tau_{\text{ext}} = 0}_{\text{rotational}} \qquad \underbrace{\sum_i f_i + f_{\text{ext}} = 0}_{\text{translational}}$$

two rows of one six-vector equation — and all of statics is in them

A brick on the ground

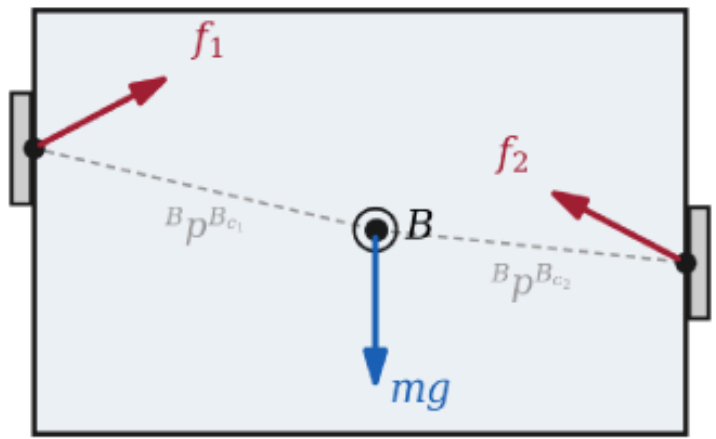


$F^B_{1,B} + F^B_{2,B} + F^B_{g,B} = 0$ six equations in 3D, three in the plane

the translational rows give $f^B_{1,B} + f^B_{2,B} = [0,\ 0,\ mg]^T$; the torque row about y makes $f^B_{1,B_z} = f^B_{2,B_z}$ when B sits midway between the contacts

class notes, Example 5.5

Holding an object still



point contacts carry no torque of their own: ${}^{B_{c_i}}\tau_{i,B} = 0$, and every moment comes from the lever arm

forces cancel mg; moments about B cancel too — or the object turns

Why statics is enough

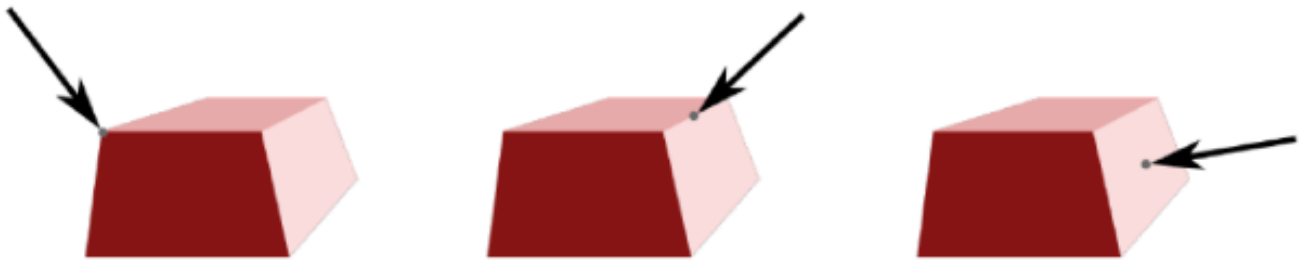
At manipulation speeds, $\|a\| \ll g$.

the inertial terms are small beside the contact forces, so the object is in equilibrium at every instant

And we are asking a *test*, not running a simulation:

can *some* admissible contact forces balance the disturbance?

What kind of contact is it?



point on vertex

point on edge

point on plane
the one we model

vertex and edge contacts we do not model

CS237B notes, Fig. 5.2

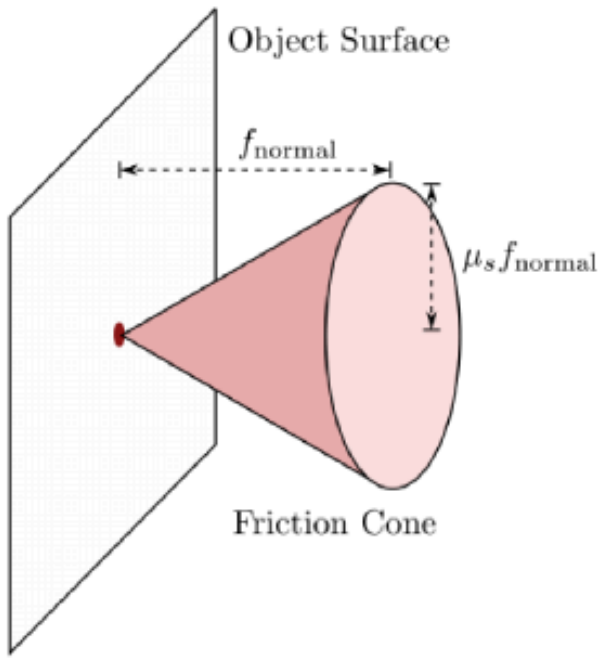
What can a contact transmit?

contact frame C with z along the inward normal — you can push, not pull: $f_{C_z}^{c_i} \geq 0$

frictionless	normal only
point with friction	$\sqrt{\left(f_{C_x}^{c_i}\right)^2 + \left(f_{C_y}^{c_i}\right)^2} \leq \mu\, f_{C_z}^{c_i}$
soft finger	$+ \ \lvert\tau_{\text{normal}}\rvert \leq \gamma\, f_{C_z}^{c_i}$

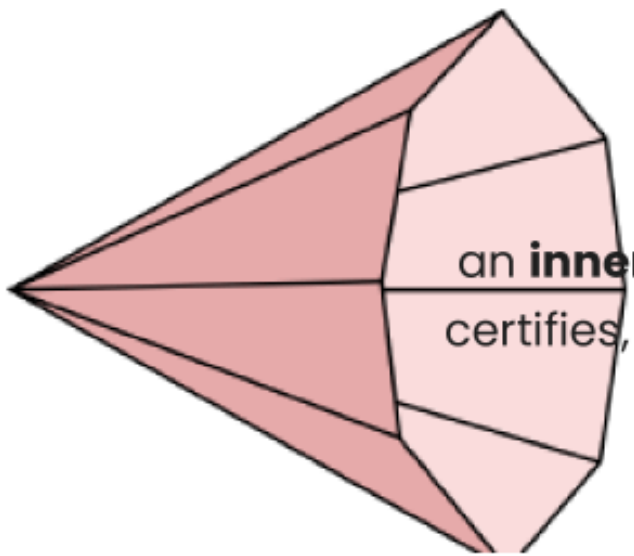
admissible forces form a **cone** $\mathcal{K}_i$: friction scales with normal force

The friction cone



half-angle arctan μ, independent of how hard you press · CS237B Fig. 5.3

Linearize it and everything becomes linear algebra



$$f_i = \sum_{j=1}^{m} \alpha_{i,j} f_{i,j}, \qquad \alpha_{i,j} \geq 0$$

an **inner** approximation: what the pyramid certifies, the true cone certifies too

CS237B notes, Fig. 5.4

Weak links

μ is not a property of a material.

a pair of materials · contamination · wear · a factor of two is unremarkable

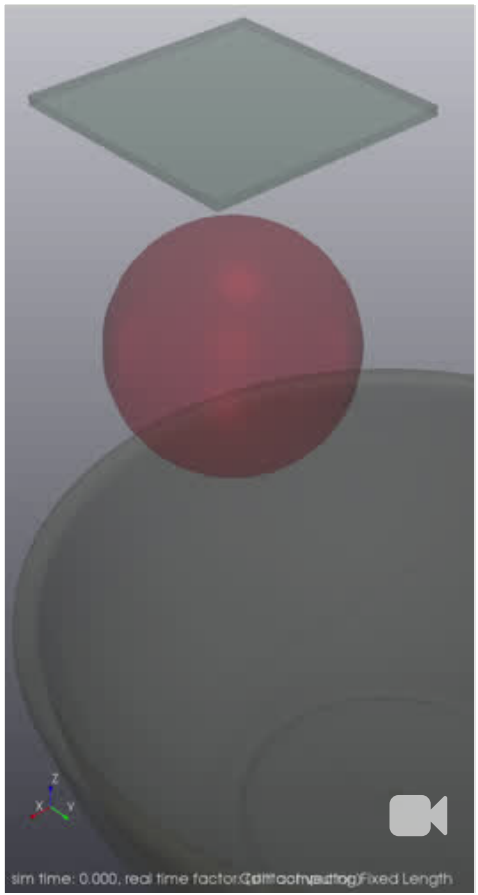
And a "point" contact is really a patch.

a soft finger flattens against the surface — the door into hydroelastic contact

Everything in Part I is exact given μ and a point.

Neither is exact in the world.

"Hydroelastic contact" as implemented in Drake

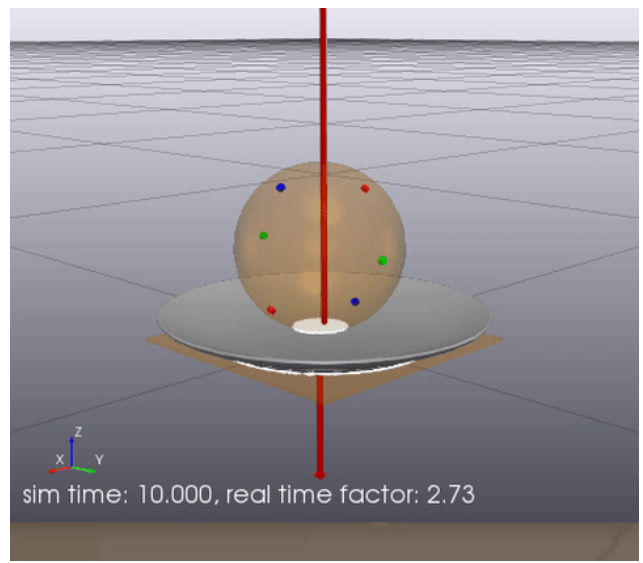


A pressure field model for fast, robust approximation of net contact force and moment between nominally rigid objects

Ryan Elandt[1,2], Evan Drumwright[1], Michael Sherman[1], Andy Ruina[2]

TOYOTA RESEARCH INSTITUTE

major contributions from Damrong Guoy, Sean Curtis, Rick Cory, Alejandro Castro, ...

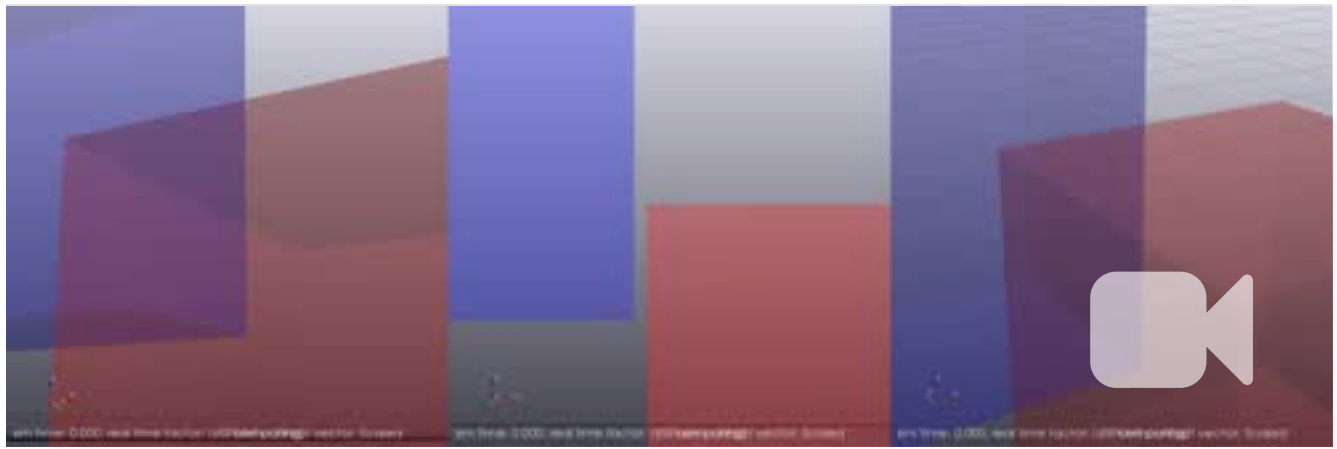


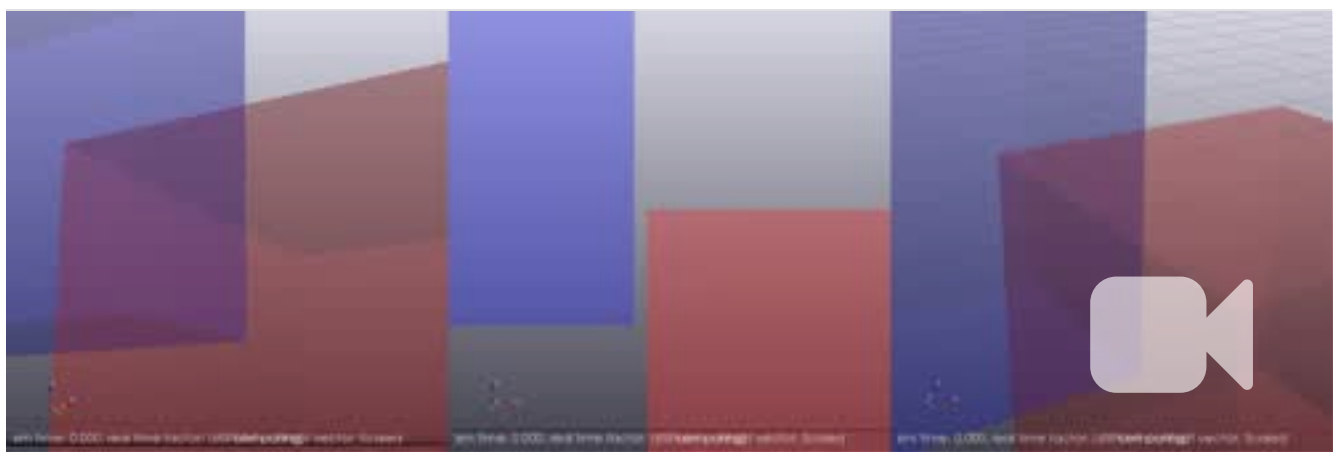
Point contact vs hydroelastic

Point contact
(discontinuous)

vs

Hydroelastic
(continuous)

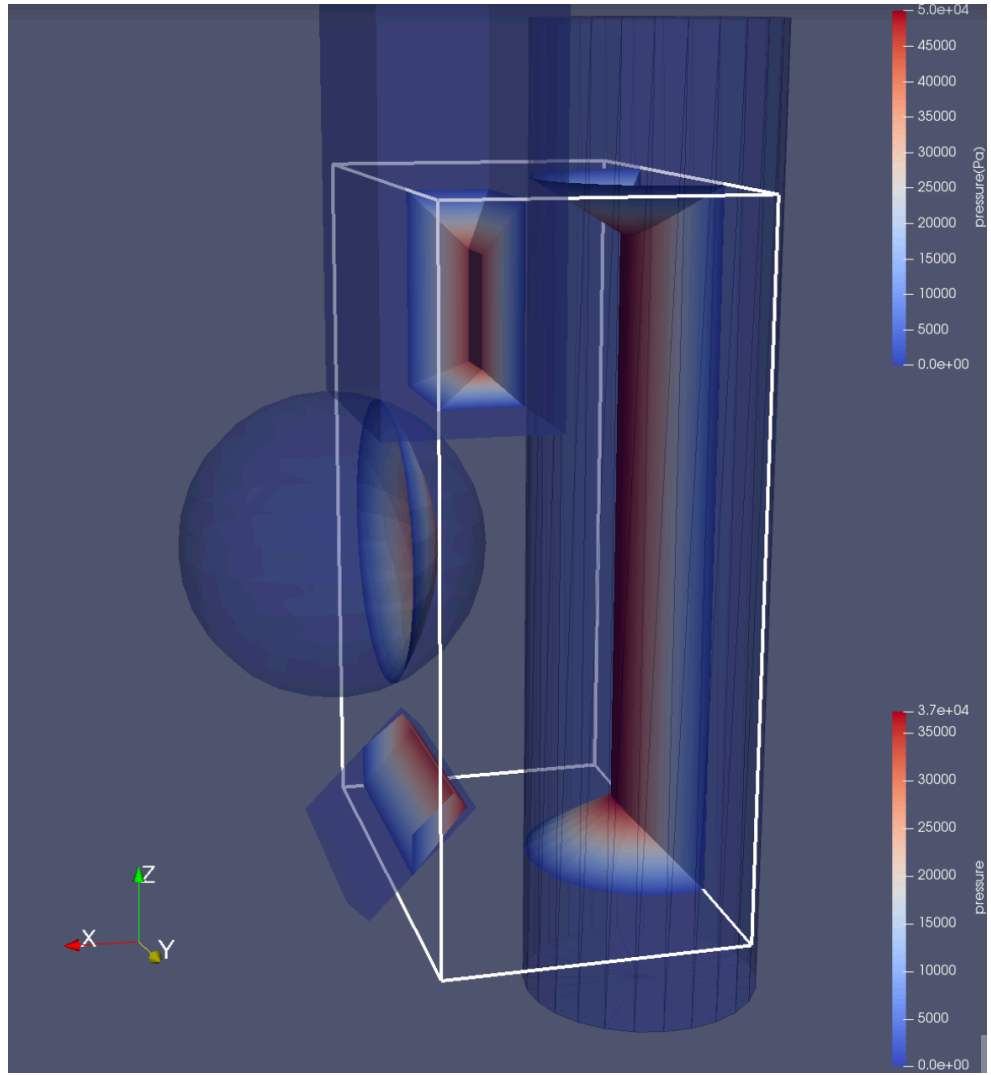




Hydroelastic is

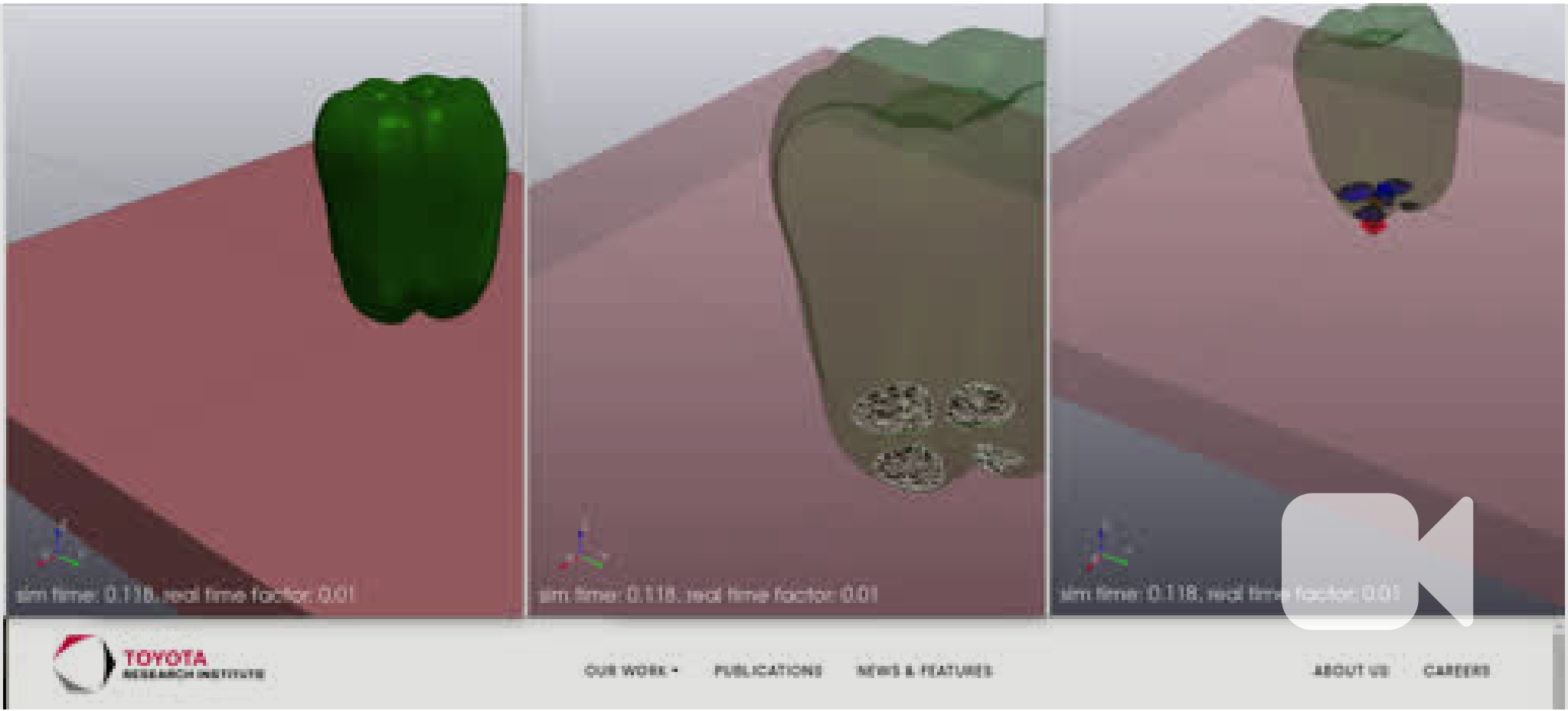
- more expensive than point contact
- (much) less expensive than finite-element models

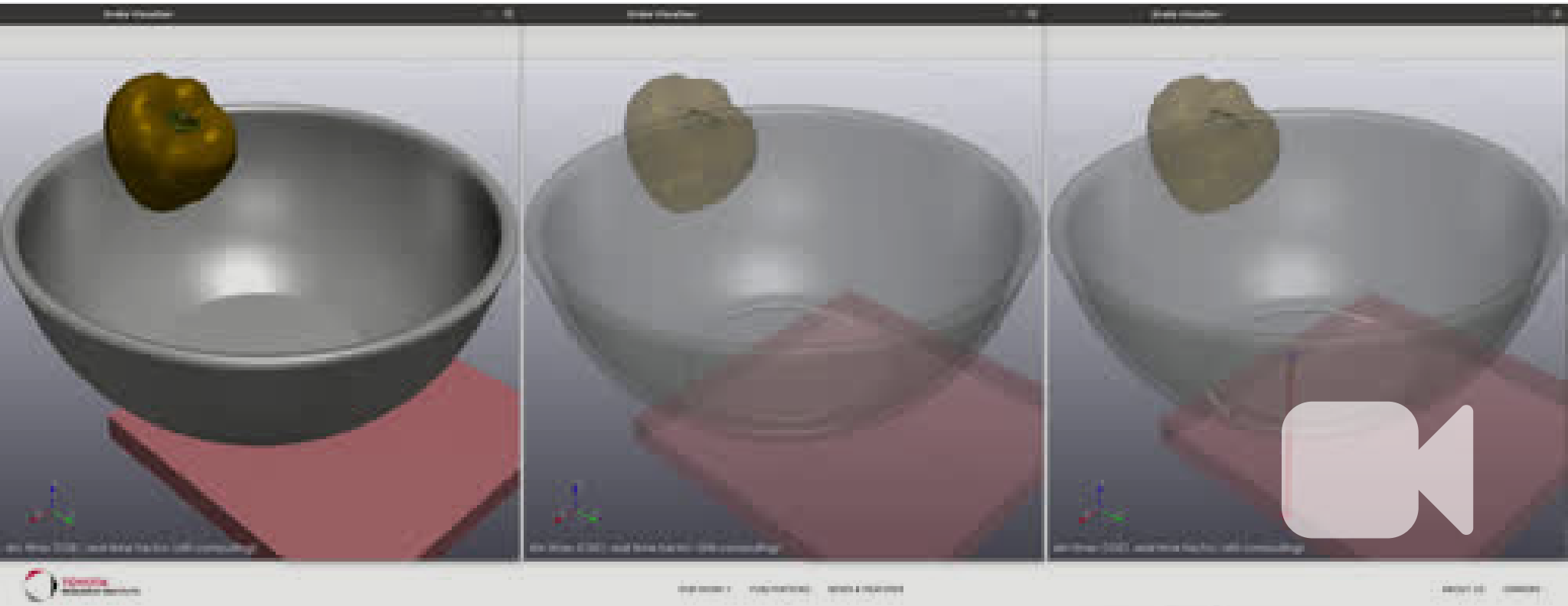
State-space (for simulation, planning, control) is the original rigid-body state.



"Hydroelastic contact" as implemented in Drake




sim time: 0.118, real time factor: 0.01
sim time: 0.118, real time factor: 0.01
sim time: 0.118, real time factor: 0.01
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The grasp map

$$G_i = \begin{bmatrix} S^{(i)} Q^{(i)} \\ Q^{(i)} \end{bmatrix} \in \mathbb{R}^{6\times 3}$$

d_i	where contact i touches, in the object frame ($S^{(i)} = S(d_i)$: the shift)
$Q^{(i)} = {}^{B}R^{C_i}$	how its contact frame is oriented (the re-express)
$f_i \in \mathbb{R}^3$	the force it applies, in its own frame — a hard finger: no moment

$$F = \sum_{i=1}^{k} G_i f_i = G \begin{bmatrix} f_1 \\ \vdots \\ f_k \end{bmatrix}, \qquad G = \begin{bmatrix} G_1 & \cdots & G_k \end{bmatrix}$$

G is *linear* and fixed by the contact locations — all the nonlinearity lives in the cones $\mathcal{K}_i$

Equilibrium, in one matrix

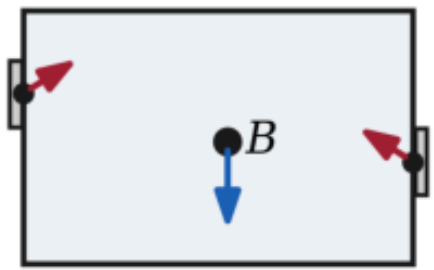
$$Gf + F^{\text{ext}} = 0$$

For which F^{ext} does an *admissible* f exist?

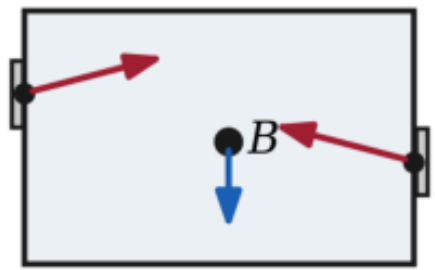
admissible $=$ inside the cones. That question is the rest of Part I.

Statics does not pin down f

same net wrench on the object — the difference is internal



light squeeze



hard squeeze

with two or more contacts G has a null space: $f + \lambda n,\ Gn = 0$, is equally in equilibrium

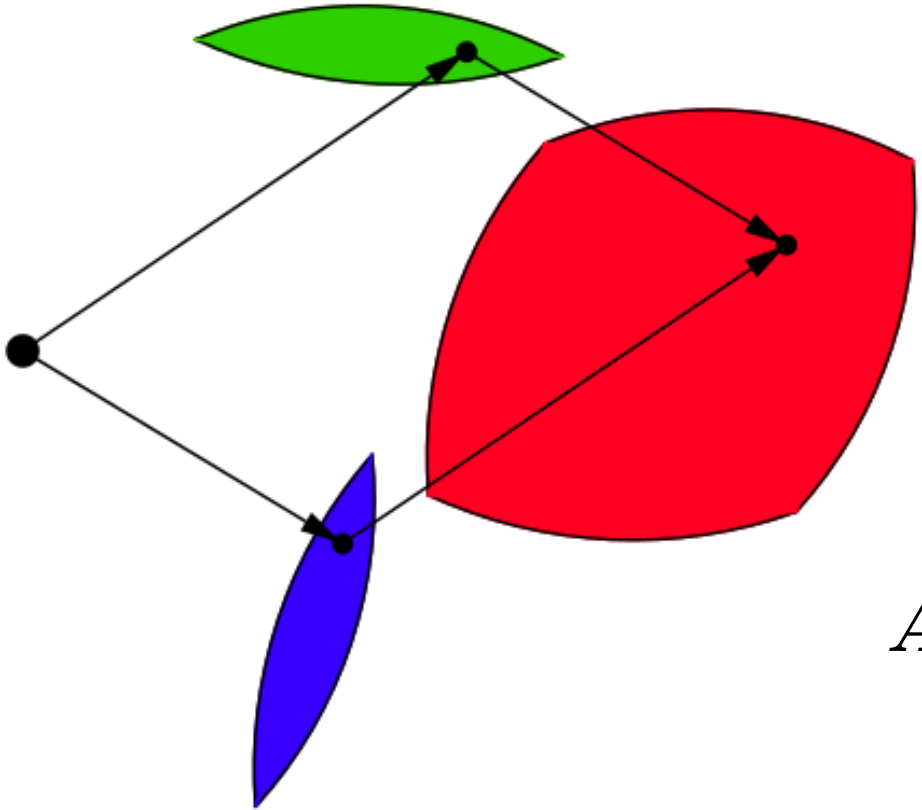
that null space is the **grip force** — free to choose, and what keeps each f_i inside its cone

Grasp wrench space

$$\mathcal{W} := \left\{ F \,\middle|\, F = \sum_{i=1}^{k} G_i f_i, \quad f_i \in \mathcal{K}_i \right\}$$

a Minkowski sum: each contact contributes a cone of wrenches

$\mathbb{R}^6$ in 3D; $\mathbb{R}^3$ in the plane (two forces, one torque)



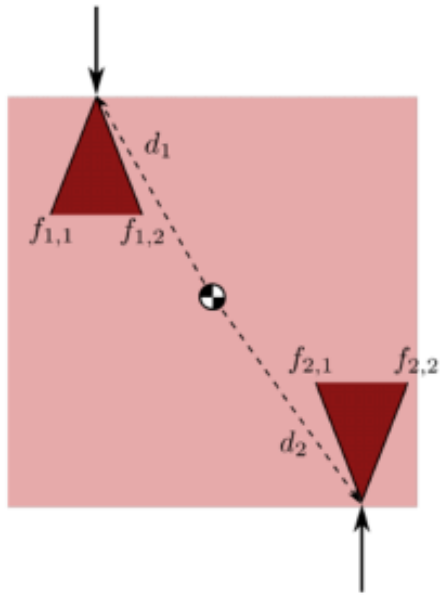
$$A \oplus B = \{a + b \mid a \in A, b \in B\}$$

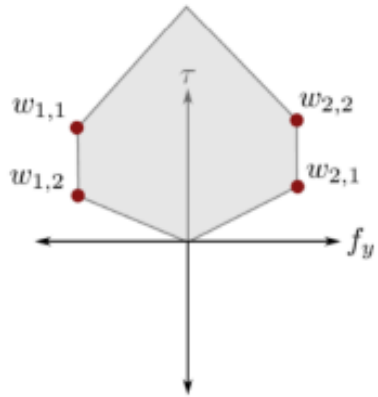
The red figure is the Minkowski sum of blue and green figures.

https://en.wikipedia.org/wiki/Minkowski_addition

https://manipulation.csail.mit.edu/clutter.html#wrench_cone_algebra

Two contacts, and what they can produce



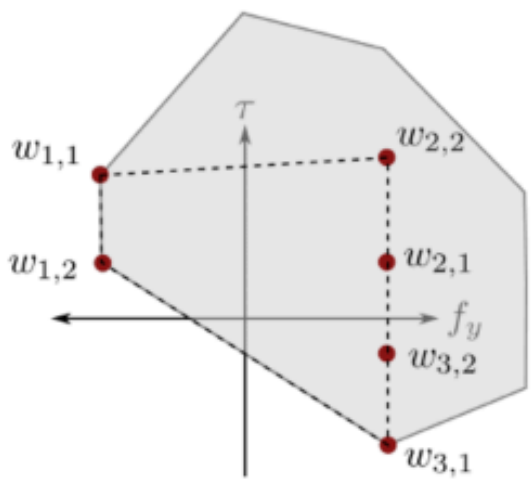


the object is in the x–y plane, so a wrench is (f_x, f_y, τ_z): this is its f_y–τ face

the source's $w_{i,j}$ is our $F_{i,j}$ — grey is $\mathcal{W}$, the sum of the two contact cones

CS237B notes, Fig. 5.6

The convex hull



one global bound $\sum_i \sum_j \alpha_{i,j} = 1$ instead of one per contact: the convex hull of the $F_{i,j}$

$$\hat{\mathcal{W}} \subseteq \mathcal{W}$$

cheap to compute, and conservative in the right direction

grey: $\mathcal{W}$, the Minkowski sum of the three contact cones. Dashed: $\hat{\mathcal{W}}$, the hull of the $F_{i,j}$ — the three-contact grasp below.

CS237B notes, Fig. 5.11

Closure

the grasp can be maintained for *every* possible disturbance load

Form closure kinematically constrained; frictionless force closure

Force closure contact forces, *relying on friction*, resist any wrench

every form-closure grasp is force closure — not conversely

precision grasps: force closure power / enveloping grasps: form closure

Force closure, precisely

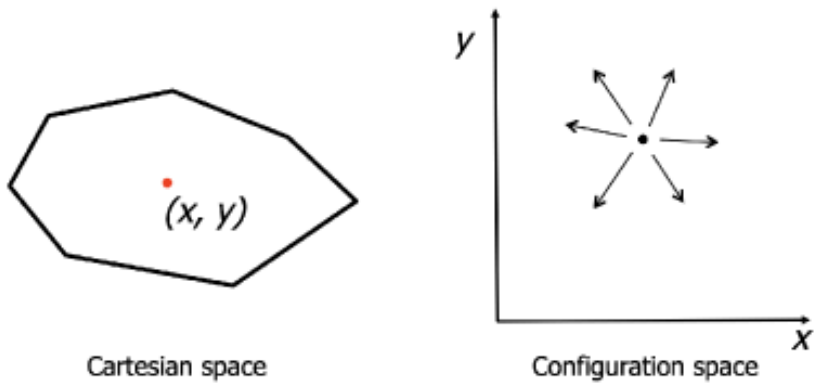
$$-F^{\text{ext}} = \sum_{i=1}^{k} G_i f_i, \quad f_i \in \mathcal{K}_i \qquad \Longleftrightarrow \qquad -F^{\text{ext}} \in \mathcal{W}$$

the origin must lie in the *interior* of $\mathcal{W}$

on the boundary there is a disturbance direction with no resistance at all

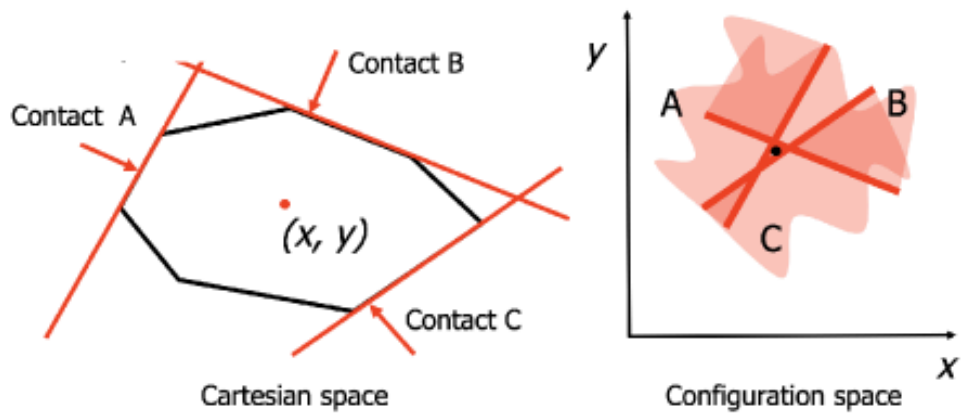
The object's motion, as a point

How many contacts needed? C-Space Analysis



Each contact removes a half-plane of motions

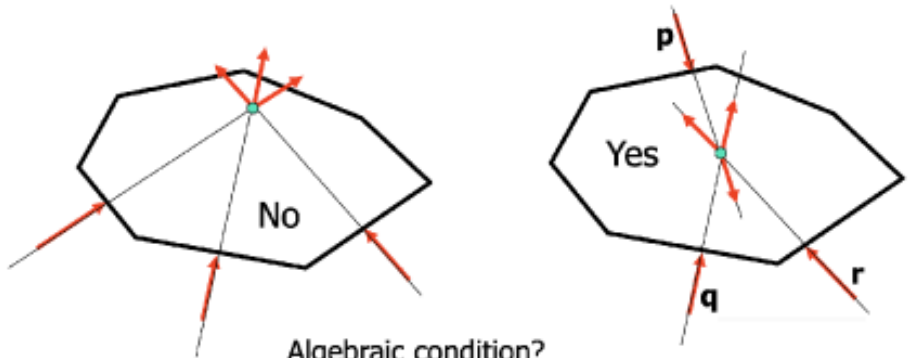
What does a point contact imply in C-Space?



Bohg, AA 274B L8, s14

Force-direction closure

Condition for Force-Direction Closure



Algebraic condition?
For force vectors **p**, **q**, **r**,
there must exist $\alpha, \beta, \gamma > 0$
s.t. $\alpha\mathbf{p} + \beta\mathbf{q} + \gamma\mathbf{r} = 0$

$$\exists\, \alpha, \beta, \gamma > 0 \quad \text{s.t.} \quad \alpha\mathbf{p} + \beta\mathbf{q} + \gamma\mathbf{r} = 0$$

sum to zero *and* every coefficient strictly positive

The general condition

$\mathcal{W} = \{F \mid F = \sum_k \beta_k F_k,\ \beta_k \geq 0\}$ fills $\mathbb{R}^n$ iff the $\{F_k\}$ contain $n+1$ vectors, n independent, and

$$\sum_{k=1}^{N} \beta_k F_k = 0 \qquad \text{for some } \beta_k > 0$$

span the space, and produce zero with every contact active

Counting contacts

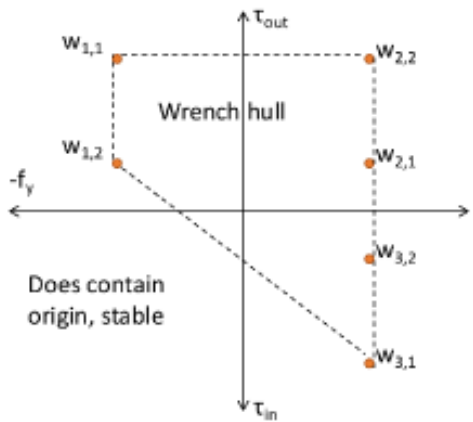
	wrenches	frictionless	with friction
2D, forces only $(n = 2)$	3	3	—
2D, forces + torque $(n = 3)$	4	4	2
3D $(n = 6)$	7	7	**3**

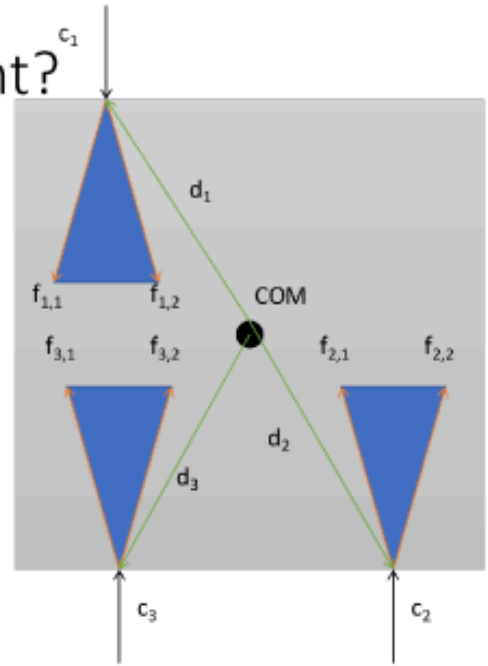
three contacts with friction suffice in 3D

seven frictionless contacts is a hand; three frictional contacts is a gripper

Why the third contact

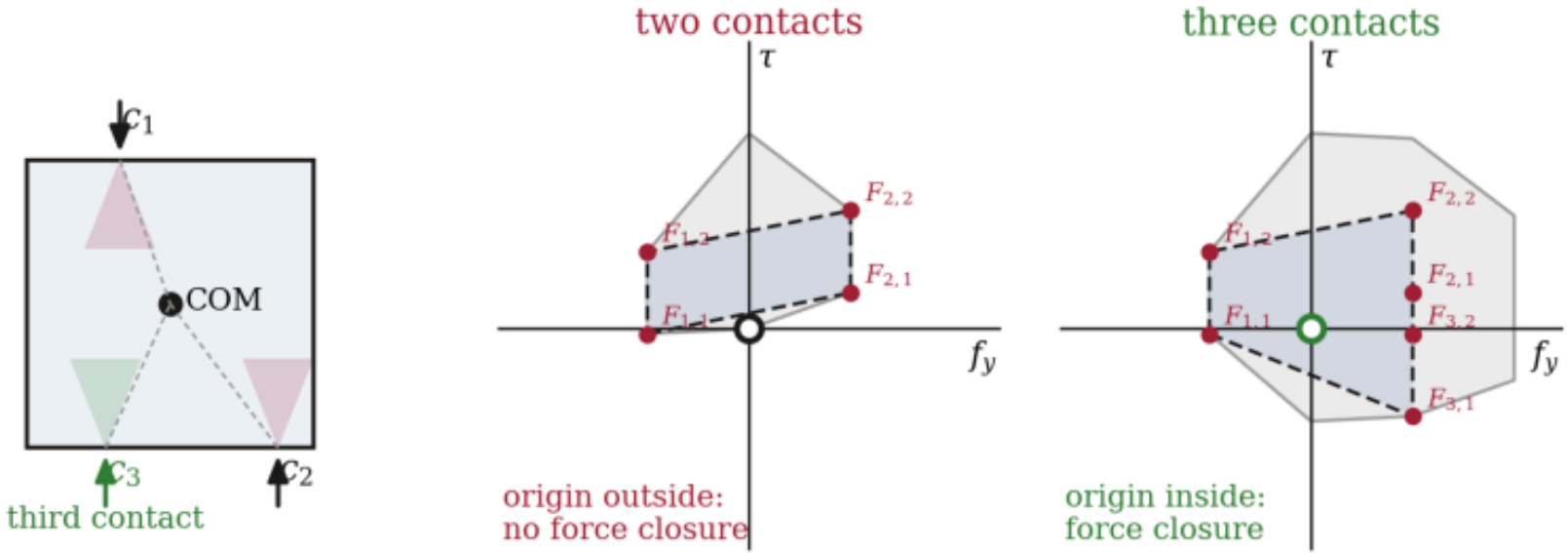
What if there was a third point?





two opposed contacts: hull on one side in torque, origin on the boundary · Bohg L8, s29

Two contacts, or three



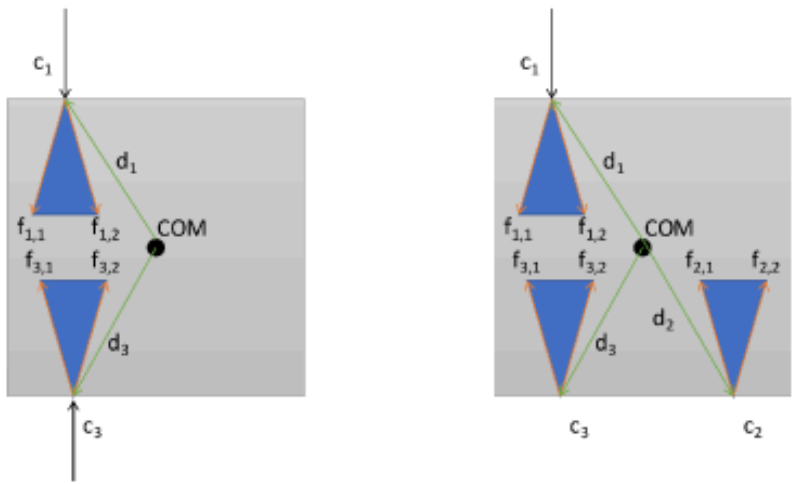
light: the wrench space $\mathcal{W}$ (the Minkowski sum) dashed: its hull $\hat{\mathcal{W}}$

with c_1 and c_2 alone every $F_{i,j}$ has positive torque about the center of mass, so the origin is a *corner* of $\mathcal{W}$, not interior to it

c_3 pulls both regions around the origin: $\sum_{i,j} \alpha_{i,j} F_{i,j} = 0$ now has a solution with every $\alpha_{i,j} > 0$

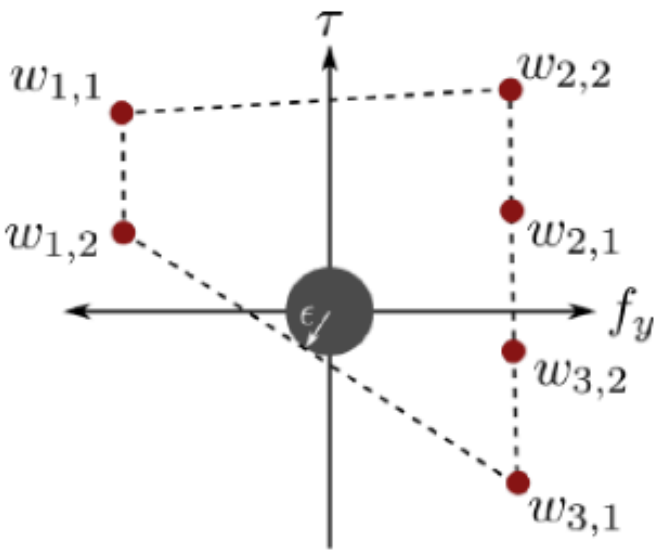
Both are in force closure

Both grasps are stable. Which one is better?



closure is a yes/no — quality is not · Bohg, AA 274B L8, s33

Worst case: the ϵ metric



radius of the largest origin-centered ball inside $\hat{\mathcal{W}}$

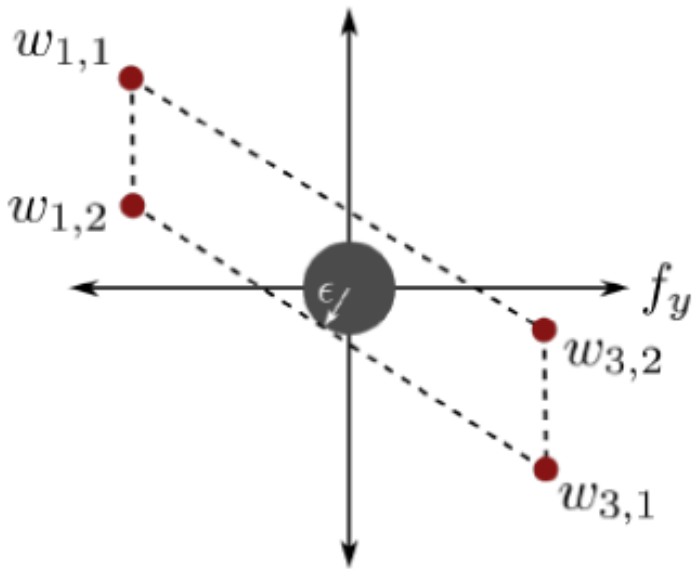
$\epsilon = 0$: not force closure at all

the touch point marks the **least resistible direction**

contact wrenches must total $1/\epsilon$ times the disturbance

CS237B notes, Fig. 5.12

Average case: the volume metric



ν = volume of the wrench hull

same worst case as the previous grasp, smaller volume

ϵ and ν can disagree — which is the reason to show both

CS237B notes, Fig. 5.13

What Part I assumed

Coulomb friction, with μ known

object geometry *and* mass distribution known — so d_i is known

a rigid body

contacts exactly where you intended

In a bin of unknown objects, seen by one depth camera: none of the four.

Optional: which forces to command

closure says *some* forces resist the disturbance; the gripper needs numbers

$$\begin{aligned} \min_{f} \quad & F_{\max} \\ \text{s.t.} \quad & f^{(i)} \in \mathcal{K}_i, \quad i = 1, \dots, M \\ & Gf + F^{\text{ext}} = 0 \end{aligned}$$

a **second-order cone program** — convex *because* the friction cones are quadratic

(no pyramid here: a solver takes the true cone directly)

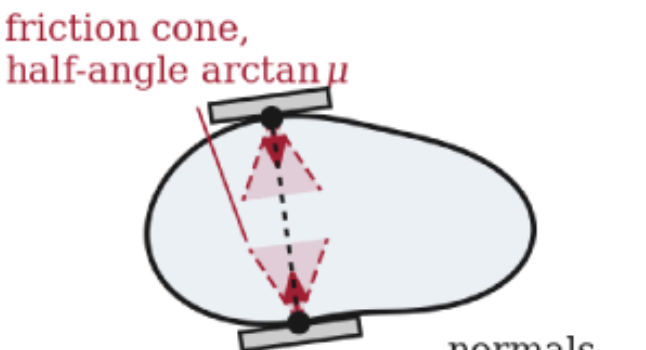
Bohg, AA 274B L8, s42–52 · Boyd & Wegbreit, T-RO 2007

Part II

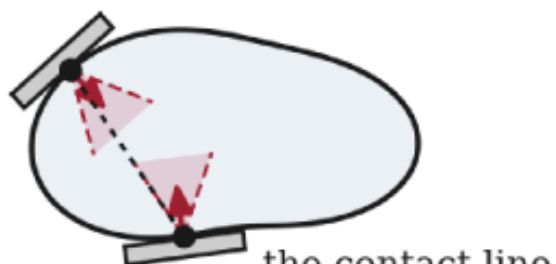
heuristic

Antipodal grasps

two contacts whose surface normals are **antiparallel** and **collinear**



antipodal



not antipodal

the cheapest geometry that gets near force closure without computing anything

Two caveats, both from Part I

Exact antipodality is not required.

the grasp survives while the contact line stays inside both cones — larger μ buys tolerance

It says nothing about torque about the grasp axis.

a long object gripped at one end is antipodal, and rotates under gravity

Normals from a point cloud

no mesh, no object identity, no center of mass — just neighbors

$$W = \sum_i (p^i - p^*)(p^i - p^*)^\mathsf{T}$$

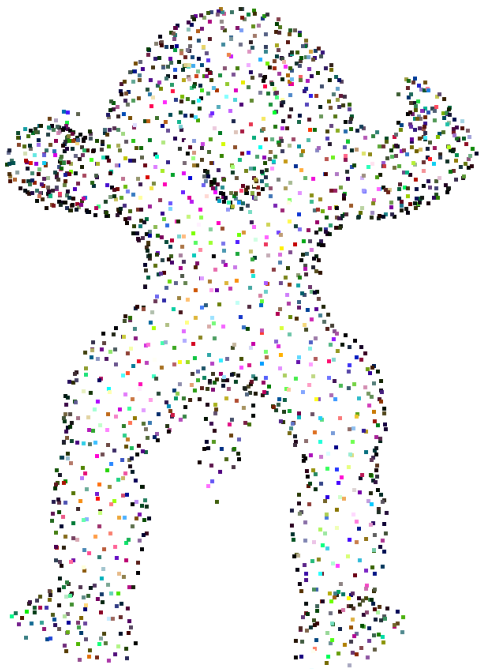
eigenvector of the **smallest** eigenvalue	the surface normal
eigenvector of the **largest** eigenvalue	the direction of least curvature

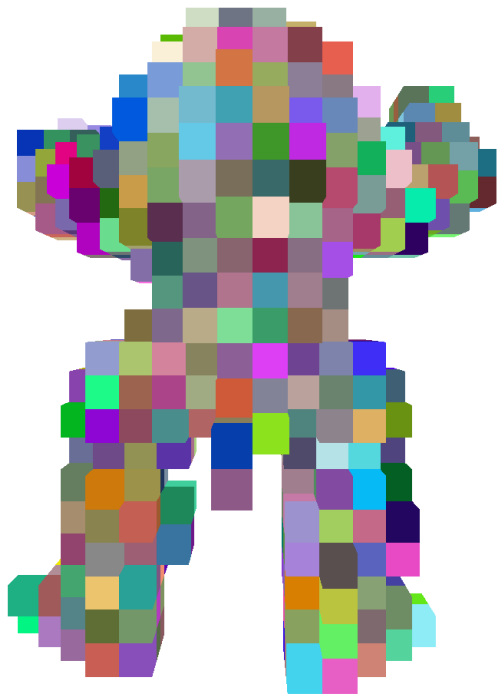
normal + curvature directions = a frame at every point, which is what proposes an *orientation*

https://manipulation.csail.mit.edu/data/mustard_bottle_point_clouds.html

https://shainarace.github.io/LinearAlgebra/pca.html

Voxelization







http://www.open3d.org/docs/latest/tutorial/Advanced/voxelization.html

Sampling antipodal grasps

1. pick a point at random
2. **ray cast into the cloud, opposite that point's normal**
3. keep the pair if the normal found is near-antipodal *and* closer than the gripper width
4. align the gripper's closing axis with the pair
5. sample orientations about it, starting from vertical; slide until the enclosed points sit centered

sample, do not optimize — then score what is between the fingers

What disappeared

no μ · no center of mass · no mesh · no closure computation

and no guarantee

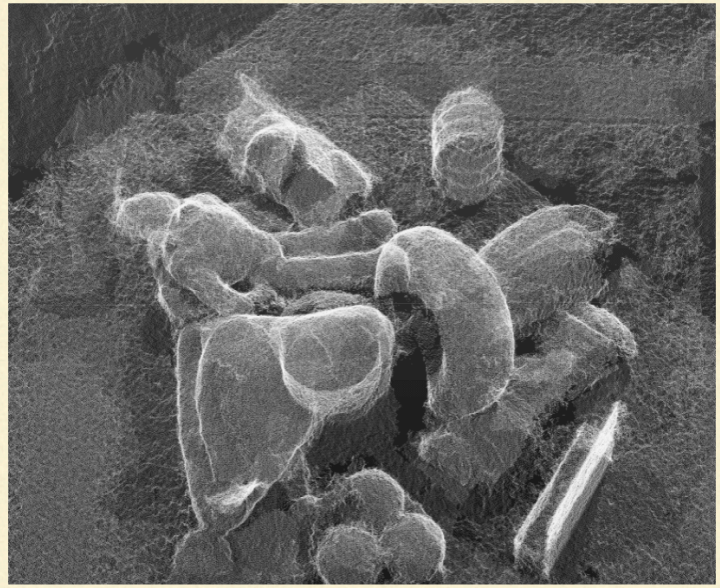
we traded the certificate for running on a bin we have never seen



(unpublished) work by Lucas Manuelli ~2017. No deep learning -- this is geometry only.

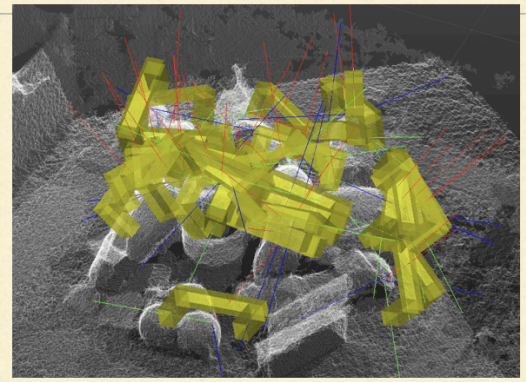
ALGORITHM: SENSOR DATA

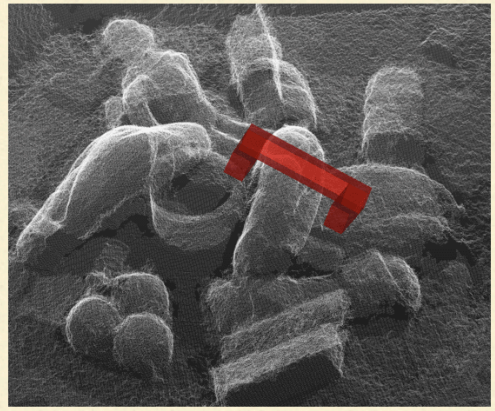
- Algorithm requires **complete pointclouds**.
- Outlier removal
- Normal estimation using camera origins
- Must be **well calibrated** (both intrinsics and extrinsics) in order to provide accurate geometry information to planner



ALGORITHM: SCORING GRASPS

- Construct a scoring function based on antipodal criterion.
- Reward grasps that have contact patch normals aligned with closing direction of gripper.
- Reward grasps that have a larger contact patch.
- In practice this always chooses sensible grasps.





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WHAT GRASPING CAN'T DO

- The presented techniques cannot
  - Identify a specific object
  - Grasp specific object
  - Place object in specific configuration
  - Manipulate object to desired configuration
  - Use a tool
  - Put away the dishes
  - and so much more!





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Part III

learned

What supplies the labels?

an analytical metric evaluated in simulation or a robot trying it

Part I against Part II, all over again

https://www.youtube.com/embed/kfe5bNt35ZI?mute=1&enablejsapi=1&rel=0

Grasp pose detection (ten Pas, Gualtieri, Saenko, Platt)

generate a large set of **6-DoF** candidates from local surface geometry: hand poses that are collision-free and enclose points between the fingers

encode the points and normals inside the closing region, as a small multi-channel image

classify each candidate good or bad with a CNN — pretrained on simulated depth from CAD models, trained on real depth scans

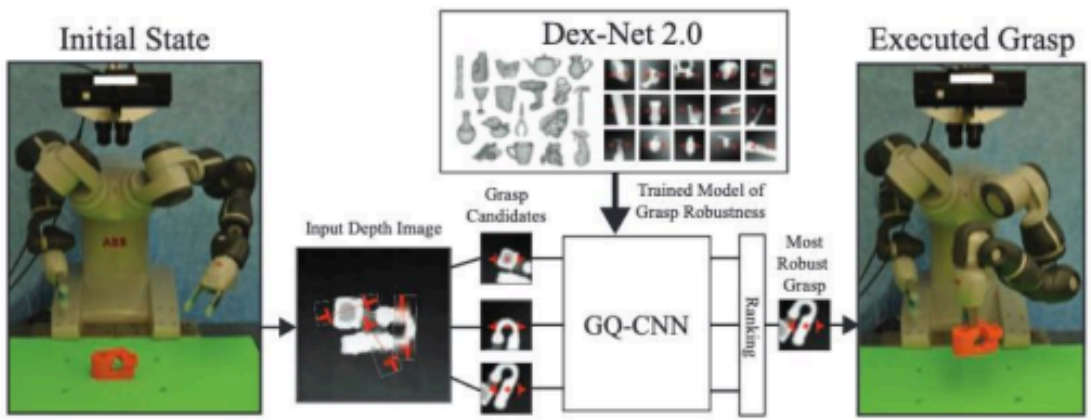
93% grasp success in dense clutter, with no object identity and no pose estimation

propose-and-score again — but the proposals are 6-DoF, not Dex-Net's top-down (i, j, φ, z)

Gualtieri et al., IROS 2016 · ten Pas et al., IJRR 2017

Dex-Net 2.0

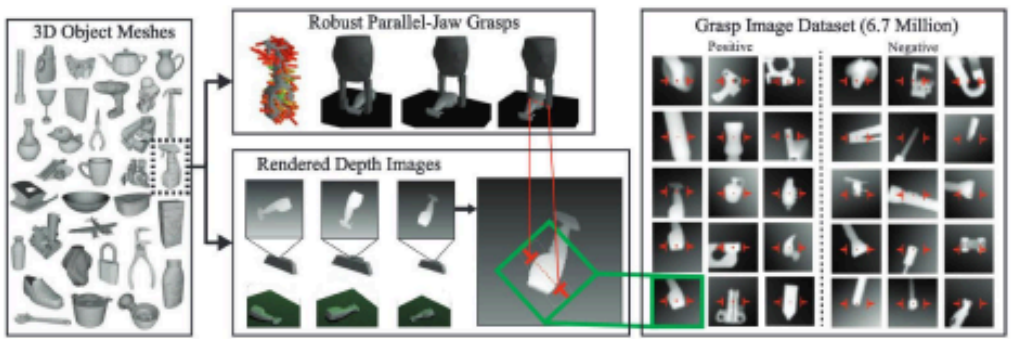
Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics



depth image → candidates → ranked by a network · Mahler et al., RSS 2017 · Bohg L9, s44

Part I as the labeling function

Dataset Generation



sample grasps on meshes → evaluate force closure under simulated uncertainty → label = fraction that hold

6.7 M labeled grasp images — and the network never sees the mesh

Where the candidates come from

on the meshes (for the dataset)

rejection sampling for **antipodal point pairs** (from Dex-Net 1.0): keep surface pairs whose normals face each other inside the friction cone — about 100 grasps per object

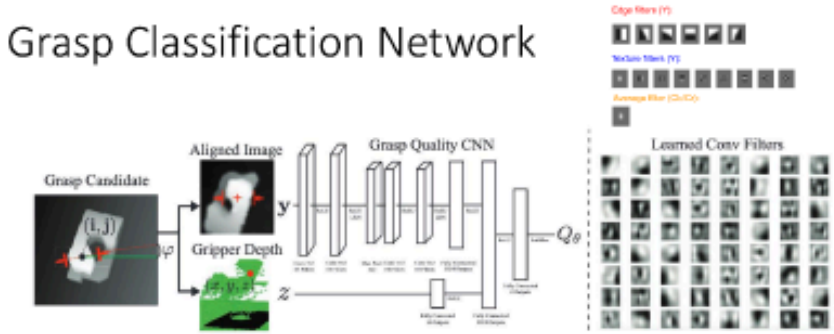
on the depth image (at run time)

normals from the **depth-image gradients**; antipodal pairs sampled **uniformly at random in image space**

$$u = (i,\ j,\ \varphi,\ z)$$

pixel, in-plane angle, gripper depth — a *planar* grasp, and exactly the patch the network is fed

GQ-CNN



$$\pi_\theta(y) = \arg\max_{u \in \mathcal{C}} Q_\theta(u, y)$$

a fast surrogate for an expensive analytical computation — trained where it is cheap, deployed where it is impossible

($\mathcal{C}$ is not a fixed list — next slide)

The policy is a search, not a sweep

draw	**100** candidates from the image sampler
score	each one with Q_θ
fit	a **3-component** Gaussian mixture to the best **25%**
draw	**50** more from that mixture
repeat	**3** iterations, then execute the best feasible grasp

the cross-entropy method — a dumb sampler, steered by the network

§12 replaces the sampler itself with something learned

The other branch

Conclusion: Two Approaches

	Dex-Net	Arm Farm
Setup	Single object in simulation	Bin of objects in real world
Number Data Points	13,000 objects, 2.5M grasps	1,100 objects, 1.7M grasps
Data Point	(object, grasp, label = probability of success)	(Image, motor command, label = ground truth success)
Diversity of Objects	Rigid, Opaque	Rigid & deformable, opaque & translucent
Object Representation	3D Mesh Model	None
Data Collection Method	Generated in simulation	Self-supervised on real hardware
Type of Learning	Deep learning, reinforcement learning	End-to-end deep learning

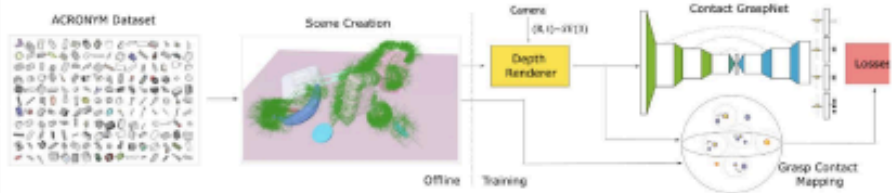
months of hardware time, no meshes, labels from outcomes · Levine et al. · Bohg L9, s66

Contact-GraspNet

Today



Contact-GraspNet: Efficient 6-DoF Grasp Generation in Cluttered Scenes. Sundermeyer et al. ICRA 2021



Simulation data is dominating





6-DoF grasps *in clutter*: the network maps observed points to grasp contacts + confidence

trained on **ACRONYM** — simulated scenes, rendered depth, simulated labels

Three systems, three questions

	question answered	mechanism
Dex-Net	how good is *this* grasp?	classify a proposed candidate
Contact-GraspNet	where is a grasp near *this* point?	regress contacts per point
GraspGen	what is the *distribution* of grasps?	sample a generative model

neither of the first two is generative: they score, they do not propose

Why generate rather than regress

Grasps are multimodal.

a mug has a rim grasp, a handle grasp, a body grasp — all good, all far apart in $SE(3)$

squared error on that target averages the modes, and the average is not a grasp

the same argument returns for diffusion policies in Lectures 8–9

GraspGen

iterative DDPM denoising in grasp-pose space, on a DiffusionTransformer

conditioned on the observed object point cloud

+ a discriminator that scores and filters the samples

on-generator training: the discriminator learns on the generator's *own* samples —

collisions and outliers it actually produces, which uniform negatives never teach

Murali et al., NVIDIA Research, arXiv:2507.13097 (2025)

Trained from what data

objects	36,366 Objaverse meshes (the LVIS-category, CC-BY subset)
grasps	**53.1 M** total, ≈ 17 M per gripper
grippers	Franka Panda, Robotiq-2f-140, 30 mm suction
sampling	2,000 grasp transforms per object, uniform around the mesh
labels	a **shaking test** in Isaac — stable contact after shaking = success

Dex-Net's label was Part I evaluated many times. This label is a physics test.

the theory has dropped out of the labeling pipeline entirely

https://graspgen.github.io/